

Khalil Lagha

Power electronics | Electrical energy systems

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4-6-month final-year internship from mid-03/2027 | One-year M2 apprenticeship from 09/2026

Apprenticeship schedule: 2-3 weeks in the company / 2-3 weeks at university

Available throughout France | Authorized to work in France

Summary

M2 CSEE student at Université Grenoble Alpes, **ranked 2nd in M1 EEA (15.43/20)**. Experience in fuel-cell modelling, converter regulation and electrical-machine control. I aim to contribute to the design and validation of electrical energy systems through verified simulation and practical engineering work.

Professional experience

Research Intern | GIPSA-lab, Grenoble

04/2026 - 07/2026 | 3.5 months

- Modelled a **10-cell proton-exchange-membrane fuel-cell (PEMFC) stack** in MATLAB/Simulink: **6% dynamic error and 1% static error**.
- Regulated voltage through a **DC-DC boost** converter using a proportional-integral (**PI**) controller, in simulation and then in practice.

Power Electronics and Automation Intern | EURL Lagha

01/2024: 15 days | 09/2024: 1 month | Two separate internships

- Worked on an industrial buck converter, **50 V to adjustable 0-48 V at 25 A**, and on **4 transformer-rectifiers rated 100 V DC / 80 A**, with ripple below 5%.
- Produced **EPLAN** schematics; contributed to a Siemens TIA Portal supervision HMI based on the supervisor's Ladder PLC program.

Projects

Indirect field-oriented control (FOC) | Python, MATLAB/Simulink

- Compared **3 implementations** of a **5 kW / 380 V induction motor**: cascaded PI and space-vector PWM (**SVPWM**) at 10 kHz; simulated speed steps settled in **52-95 ms with no overshoot**.

Power-grid analysis | MATLAB, individual project

- Compared **3 load-flow methods** on **3-118-bus networks**, with PQ/PV/slack buses; computed impedance/admittance matrices and studied transient stability.

Education

M2 CSEE - Electrical Energy Systems Design | Université Grenoble Alpes

09/2026 - 08/2027 | In progress

M1 EEA (EECS) | Université Grenoble Alpes

2025 - 2026 | Ranked 2nd in the cohort, 15.43/20

Electrical Engineering cycle | École Nationale Polytechnique, Algiers

2021 - 2025 | 4 of 5 years completed

Skills

Internship practice: MATLAB/Simulink, buck/boost converters, PI regulation, rectification, EPLAN, contribution to a TIA Portal HMI.

Projects: Python, FOC/SVPWM, DC-motor cascade control, power-grid analysis, MATLAB neural networks.

Knowledge: electrical machines, transformers, LQR, observers, C/C++ and real-time embedded systems.

Languages

French: C1 | English: C2

Certifications

SLB: NEST and SIPP Level 2, industrial-operations safety, 07/2025